

IDBP Phase 2 — Project Module Specification

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1. Overview

The **Project Management Module** is a sub-module of IDBP Phase 2. It manages construction/infrastructure work orders across multiple states and departments. The module maintains a **Master Data File** as the single source of truth for all projects, and downstream report tables (e.g., Fund Reports) reference it via a foreign key.

2. Roles & Access Scope

Role	Description
Admin	Highest privilege. Full access to Master Data and all reports. Only role that can change project status, view/restore soft-deleted records, and access the Master Data sheet in the UI.

Role	Description
Project Manager	Staff role under Admin. Scoped to the Project Management Module only. Can create and edit Fund Reports. No access to Master Data. Project Manager scope is currently universal across all projects (no per-project assignment).

Note: Additional roles may be introduced in future phases.

3. Master Data File

3.1 Schema

Table name: `projects_master`

Naming rationale: `master_data` is intentionally avoided — it is too generic and will collide semantically with future module tables such as `vendors`, `customers`, `employees`, and `products`, all of which are equally "master data". `projects_master` is scoped and unambiguous.

Column	Type	Constraints	Notes
<code>work_order_no</code>	<code>VARCHAR</code>	PRIMARY KEY, NOT NULL, IMMUTABLE	e.g., <code>WB_APD_101</code> . Cannot be edited after creation by anyone including Admin.
<code>estimate_no</code>	<code>VARCHAR</code>	NOT NULL	
<code>site_details</code>	<code>TEXT</code>	NOT NULL	
<code>state</code>	<code>VARCHAR</code>	NOT NULL	
<code>district</code>	<code>VARCHAR</code>	NOT NULL	
<code>zone</code>	<code>VARCHAR</code>	NOT NULL	
<code>department</code>	<code>VARCHAR</code>	NOT NULL	Options: <code>PWD</code> , <code>PHE</code> , <code>Irrigation</code> , <code>WRDD</code> (extend as needed)
<code>status</code>	<code>ENUM</code>	NOT NULL, DEFAULT <code>'Running'</code>	Allowed values: <code>Running</code> , <code>Closed</code> , <code>Under Maintenance</code>
<code>edited_by</code>	<code>VARCHAR</code>	NOT NULL	Stored as user mobile number (e.g., <code>+918276071523</code>)
<code>edited_at</code>	<code>TIMESTAMPZ</code>	NOT NULL	Stored in IST (GMT+5:30)

PK/FK relationship: `work_order_no` is the Primary Key of `projects_master` and serves as the Foreign Key in all downstream report/transaction tables (e.g., Fund Reports).

Immutability: `work_order_no` must be enforced as immutable at both the application layer (disable field post-creation) and ideally at the database layer (trigger or policy).

No physical deletion: Master Data records are never deleted. Projects are retired via status changes only.

3.2 Seed Data

The following 20 records represent the initial dataset:

work_order_no	estimate_no	site_details	state	district	zone	department
WB_APD_101	APD_1	Main Road Work-1	West Bengal	Alipurduar	North Bengal	PWD
WB_BAN_102	BAN_2	Main Road Work-2	West Bengal	Bankura	South Bengal	PWD
WB_BIR_103	BIR_3	Main Road Work-3	West Bengal	Birbhum	South Bengal	PWD
WB_COB_104	COB_4	Main Road Work-4	West Bengal	Cooch Behar	North Bengal	PWD
SK_GAN_201	GAN_1	Main Road Work-5	Sikkim	Gangtok	East Sikkim	PWD
SK_GYA_202	GYA_2	Pipe Line Work-1	Sikkim	Gyalshing	West Sikkim	PHE
OD_ANG_301	ANG_1	Pipe Line Work-2	Odisha	Angul	Central Odisha	PHE
OD_BAL_302	BAL_2	Pipe Line Work-3	Odisha	Balangir	Western Odisha	PHE
OD_BALS_303	BALS_3	Pipe Line Work-4	Odisha	Balasore	North Odisha	PHE
JH_BOK_401	BOK_1	Pipe Line Work-5	Jharkhand	Bokaro	North Jharkhand	PHE
JH_CHA_402	CHA_1	River Embankment Construction- 1	Jharkhand	Chatra	North Jharkhand	Irrigation
JH_DEO_403	DEO_1	River Embankment Construction- 2	Jharkhand	Deoghar	East Jharkhand	Irrigation
BH_ARA_501	ARA_1	River Embankment Construction- 3	Bihar	Araria	North Bihar	Irrigation

work_order_no	estimate_no	site_details	state	district	zone	departmer
BH_ARW_502	ARW_1	River Embankment Construction- 4	Bihar	Arwal	South Bihar	Irrigation
BH_AUR_503	AUR_1	River Embankment Construction- 5	Bihar	Aurangabad	South Bihar	Irrigation
BH_BAN_504	BAN_1	Agriculture Irrigation Work-1	Bihar	Banka	South Bihar	WRDD
BH_BEG_505	BEG_1	Agriculture Irrigation Work-2	Bihar	Begusarai	North Bihar	WRDD
WB_PUM_601	PUM_1	Agriculture Irrigation Work-3	West Bengal	Purba Medinipur	South Bengal	WRDD
WB_PUR_602	PUR_1	Agriculture Irrigation Work-4	West Bengal	Purulia	South Bengal	WRDD
WB_S24_603	S24_1	Agriculture Irrigation Work-5	West Bengal	South 24 Parganas	South Bengal	WRDD

3.3 Business Rules

- `work_order_no` cannot be edited after creation. Enforce at application + DB layer.
- The `status` field renders as a **dropdown** in the Admin UI with exactly three options: `Running`, `Closed`, `Under Maintenance`.
- Only Admin can change project status.
- Master Data records cannot be physically deleted under any circumstance through the application UI.
- The Master Data sheet link/menu item is only visible to Admin users in the Project Module navigation.
- All modifications to Master Data must be logged in the system audit log.

4. Project Status & Permissions Matrix

The `status` of a `work_order_no` in the Master Data table governs mutability of all downstream reports referencing that work order.

Status	Create Reports	Edit Reports	View Reports
Running	✔ Yes	✔ Yes	✔ Yes
Closed	✗ No	✗ No	✔ Yes
Under Maintenance	✔ Yes	✔ Yes	✔ Yes

Enforcement: When `status = 'Closed'`, all report rows linked to that `work_order_no` become **immutable** for all users including Admin. The only way to resume editing is for Admin to explicitly change the status back to `Running` or `Under Maintenance` in the Master Data sheet.

5. Feature Permissions Matrix

Feature	Admin	Project Manager
View Master Data Sheet	✔	✗
Edit Master Data Sheet	✔	✗
Change Project Status	✔	✗
Create Fund Report	✔	✔
Edit Fund Report	✔	✔
Delete Fund Report (soft)	✔	✗
View Soft-Deleted Records	✔	✗
Restore Soft-Deleted Records	✔	✗
Bulk Import via Excel	📅 Planned	✗

6. Fund Reports (FK-Linked Tables)

Note: Detailed Fund Report schema to be specified separately. This section defines the integration contract with Master Data.

6.1 Foreign Key Relationship

- All report/transaction tables (including Fund Reports) must have their **own independent Primary Key**, separate from `work_order_no`. Example: `fund_report_id` UUID PRIMARY KEY. This is required because one project will have many fund reports over its lifetime — `work_order_no` alone cannot uniquely identify a report row.
- All report/transaction tables must also have a `work_order_no` column as a **Foreign Key** referencing `projects_master.work_order_no`.

6.2 Auto-fill Behaviour

- When a user enters or selects a `work_order_no` in a Fund Report form, the following fields **must be auto-populated** from the Master Data record in real time:
 - `estimate_no`
 - `site_details`
 - `state`
 - `district`
 - `zone`
 - `department`
- These auto-filled fields are **display-only** in the form — they are not stored in the Fund Report table.

Storage rule: Store `work_order_no` only. Fetch all other Master Data columns live via join/lookup at read time. Do not duplicate `projects_master` columns in report tables.

6.3 Mutability Gate (Status Check)

Before allowing create or edit on any report, the application must check the `status` of the linked `work_order_no` in Master Data and enforce the matrix defined in [Section 4](#).

7. Soft Deletion Policy

Applies to **all** report and transaction tables (Fund Reports and any future tables).

7.1 Soft Delete Fields

Every report/transaction table must include the following system-managed columns:

Column	Type	Notes
is_deleted	BOOLEAN	Default FALSE
deleted_by	VARCHAR	User ID / mobile number of the actor
deleted_at	TIMESTAMPTZ	Timestamp in IST (GMT+5:30)

7.2 Behaviour

- Soft-deleted records have `is_deleted = TRUE`.
- Normal application views (for all roles) **filter out** `is_deleted = TRUE` records.
- Admin users can view soft-deleted records through a dedicated admin interface.
- Admin users can **restore** soft-deleted records. Restoration resets `is_deleted = FALSE` and clears `deleted_by` / `deleted_at` (or logs a restoration event in the audit log).
- Project Managers **cannot** restore or permanently remove soft-deleted records.
- The application UI must **never** provide a permanent delete option for any business record.
- Physical deletion (if ever required) is strictly a database-level admin operation and is outside the application's scope.

8. Audit Log

8.1 Triggering Events

The audit log must record an entry for each of the following actions:

- Record Creation
- Record Modification
- Project Status Change
- Soft Deletion
- Record Restoration

8.2 Audit Log Entry Schema

Table name (suggested): `audit_log`

Column	Type	Notes
<code>id</code>	<code>BIGSERIAL</code> / UUID	Auto-generated PK
<code>user_id</code>	<code>VARCHAR</code>	Mobile number of the acting user
<code>action</code>	<code>VARCHAR</code>	e.g., <code>CREATE</code> , <code>EDIT</code> , <code>STATUS_CHANGE</code> , <code>SOFT_DELETE</code> , <code>RESTORE</code>
<code>module_name</code>	<code>VARCHAR</code>	e.g., <code>Project Management</code> , <code>Fund Report</code>
<code>record_identifier</code>	<code>VARCHAR</code>	Primary key of the affected record (e.g., <code>work_order_no</code>) or report ID)
<code>old_value</code>	<code>JSONB</code>	<code>NULL</code> on CREATE. Snapshot of changed fields before the operation.
<code>new_value</code>	<code>JSONB</code>	<code>NULL</code> on SOFT_DELETE/RESTORE (no field changes). Snapshot of changed fields after the operation.
<code>timestamp</code>	<code>TIMESTAMPTZ</code>	IST (GMT+5:30)

Why `old_value` / `new_value`: Knowing that a status changed is not enough for audit purposes. Capturing the before/after state answers "changed from what to what".

Example entry for a status change:

```
json
{
  "old_value": { "status": "Running" },
  "new_value": { "status": "Closed" }
}
```

Only the modified fields need to be captured, not the full row snapshot.

9. Standard Audit Fields (All Tables)

Every report and transaction table in the system must carry the following standard fields in addition to their own business columns:

Column	Type	Notes
<code>created_by</code>	<code>VARCHAR</code>	User ID / mobile number
<code>created_at</code>	<code>TIMESTAMPTZ</code>	IST (GMT+5:30)
<code>edited_by</code>	<code>VARCHAR</code>	User ID / mobile number
<code>edited_at</code>	<code>TIMESTAMPTZ</code>	IST (GMT+5:30)
<code>is_deleted</code>	<code>BOOLEAN</code>	Default <code>FALSE</code>
<code>deleted_by</code>	<code>VARCHAR</code>	Nullable; set on soft delete
<code>deleted_at</code>	<code>TIMESTAMPTZ</code>	Nullable; set on soft delete

10. Data Import

Method	Status
Manual entry via Admin UI	 Current implementation
Bulk Excel import (Admin only)	 Planned for a later phase

11. Implementation Notes & Constraints

#	Constraint
1	<code>work_order_no</code> is immutable post-creation. Enforce at application layer (disable input) and database layer (trigger or row-level policy).
2	<code>projects_master</code> rows are never physically deleted through the application. No DELETE endpoint should exist for <code>projects_master</code> .
3	All report tables store <code>work_order_no</code> only; all other <code>projects_master</code> columns are fetched live via FK join. No denormalization.
4	Status check must be performed server-side before any create/edit operation on a report, not just client-side.
5	All timestamps must be stored and displayed in IST (UTC+5:30).

Constraint

- 6 `edited_by` / `created_by` values are stored as the user's mobile number (e.g., `+918276071523`).
 - 7 The Master Data sheet menu item must be conditionally rendered — visible only to Admin role.
 - 8 The `status` dropdown in the Master Data UI must only expose: `Running`, `Closed`, `Under Maintenance`.
 - 9 Soft-deleted records must be invisible in all default queries. Use a default filter `WHERE is_deleted = FALSE` or a database view.
 - 10 The audit log is append-only. No update or delete operations should be permitted on `audit_log`.
 - 11 **Concurrency (last-write-wins):** In the current phase, the last successful save operation wins and overwrites `edited_by` / `edited_at`. No optimistic locking or conflict detection is implemented. Record locking or versioning mechanisms may be introduced in a future phase.
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End of Specification — IDBP Phase 2, Project Management Module v1.1